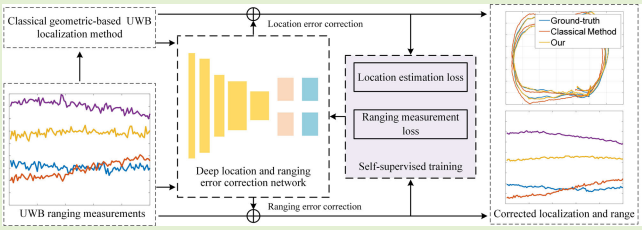
Self-Supervised Deep Location and Ranging Error Correction for UWB Localization

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***Abstract*—In this article, we propose a self-supervised deep network that mainly applies location and ranging error corrections to a classical ultrawideband (UWB) localization approach to improve its accuracy. The core of this method is the self-supervised learning strategy which removes the requirement for ground truth in the training process and thus reduces the training cost of the method. To this end, we first use an existing classical UWB localization approach to provide an initial tag location, and then build a deep location and ranging error correction (DLRC) network to jointly estimate the tag position corrections and distance corrections. The self-supervised training strategy is built based on these regressed corrections and the initial tag position as well as the fixed anchor positions and raw ranging measurements, through the topological structure of the UWB sensors. Finally, the initial tag location can be corrected by the trained DLRC network. This proposed method makes use of high-level representations of the UWB measurements by the deep network, while the geometric information important for UWB localization is also considered through the self-training strategy and the classical approach. Therefore, it can effectively improve the localization performance compared to the state-of-the-art classical approach. This improvement is also verified by conducting real-world experiments.**



***Index Terms*—Indoor localization, location and ranging error correction, self-supervised, ultrawideband (UWB) sensors.**

I. Introduction

ULTRAWIDEBAND (UWB) sensors are widely used for pedestrian or robot localization owing to their high precision measurements [1], [2], [3]. Generally, the UWB tag is placed on a carrier. Meanwhile, at least four UWB anchors are installed in a scene with known and fixed positions. The distances between the tag and each anchor can be measured based on the time of arrival (TOA). Then, the 3-D positions of tag can be computed with these ranging measurements [4].

The classical UWB localization approaches are commonly based on least squares [5], filter [6], [7], or optimization strategies [8], [9]. These approaches can make full use of the topological structure of the UWB sensors. Besides, some carrier motion constraints (e.g., the maximum velocity of the carriers) are also used to compute accurate localization results. However, they suffer from the precision decline in complex environments or are caused by the limited structure of anchors. For example, when the height difference between anchors is smaller than the coordinate differences of x-axis or y-axis, a larger localization error in z-axis will be provided.

Recently, with the deep learning successful applications in the fields of image classification [10] and natural language processing [11], it also achieves considerable progress in deep learning-based UWB localization task [12], [13], [14], [15]. In these methods, a deep network is constructed and trained as an estimator to compute the tag location from the UWB ranging measurements. Compared to the classical approach, the advantages of deep learning-based method are manifested in two aspects: 1) it can extract the deep features from the UWB measurements, yielding accurate regression of tag location and 2) the training process allows the network to find appropriate parameters to build accurate relationship between the UWB measurements and estimated values. Thus, accurate results can be obtained even in a complex

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scene. However, these methods also have limitations as follows.

Existing methods generally precollect the ground-truth positions and use them to train the model in a supervised manner. Owing to the existing ground truth, the deep network can suppress the measurement noise. Besides, this process is also not influenced by the structure of UWB sensors. However, it is costly and inconvenient for the localization system to collect the ground truth in practice, since additional sensors with a higher accuracy and cost such as a 3-D LiDAR or a motion capture system are needed. This requirement will increase the cost of the training process and hinder the application of the supervised learning-based methods.

Therefore, in this article, we propose a self-supervised learning strategy for the designed deep network. The core of this self-supervised training strategy is to consider the geometric information of the UWB sensors to overcome the lack of the ground truth. Specifically, we use the classical approach to provide an initial tag location. Meanwhile, we design a deep location and ranging error correction (DLRC) network that estimates corrections of the initial tag location and corrections of raw ranging measurements simultaneously. By using the topological information of the UWB sensors, two losses, namely location estimation loss and ranging measurements loss, can be defined from the initial tag locations, location corrections, ranging error corrections, anchor positions, and raw ranging measurements. Through minimizing these two losses together, the proposed DLRC network can be trained without the ground truth. After the self-training process, the initial tag location and the measured distance can be corrected by the DLRC network.

Thus, our proposed UWB localization method enjoys two advantages: 1) it can utilize high-level deep features of the ranging measurement and avoid the need for ground truth for training and 2) it can estimate accurate tag locations and distances between tag and anchors jointly. In addition, in our method, the topological information of the UWB sensors is exploited via the introduction of the classical approach and the training strategy.

The main contributions of this article are summarized as follows.

1) A self-supervised training strategy is proposed to train the proposed DLRC network without involving ground truth. To the best of our knowledge, it is the first self-supervised learning scheme for a deep learning-based UWB localization system.
2) The DLRC network is designed to correct the localization results for classical methods as well as the ranging measurements. This deep learning-based framework can extract the high-level spatial features from the UWB measurements, thus achieving high-accuracy localization performance.
3) The real-world experiments in different scenes with different UWB devices are conducted to demonstrate the improvement of our proposed method in comparison to the classical approaches.

The rest of this article is organized as follows. After briefly reviewing the related works of our method in Section II, we detail our proposed DLRC network and self-supervised training strategy in Section III. In Section IV, we give a comprehensive evaluation of our method, while the conclusion is given in Section V.

II. Related Work

In terms of the UWB localization methods with the distance information, the most commonly used classical UWB localization approach is based on the least squares [5]. For this approach, the equations can be built based on the ranging measurements and the known anchor positions as follows:

$$\left\| \widehat{\boldsymbol{p}}_t - \boldsymbol{A}_i \right\|_2 = \hat{d}_{t,i} \tag{1}$$

where $\widehat{\boldsymbol{p}}_t$ is the tag location to be estimated, $\boldsymbol{A}_i$ is the position of the ith anchor, and $\hat{d}_{t,i}$ is the measured distance between the tag and the ith anchor. Then, an analytical solution of these equations can be obtained by the least squares method. This approach makes full use of the topological structure of the UWB sensors to compute the tag positions. However, it is not sufficiently robust to the measurement error. Besides, the matrix inversion operation in the calculation process of the least squares method is prone to the unfriendly topological structure of the UWB sensors, leading to a large localization error in limited situations.

The filter-based approach [6], [7] and the optimization-based approach [8], [9] are proposed to overcome these issues. The tag velocity [7], [9] or the bearing information [6] is introduced in these approaches to suppress the negative influence of ranging errors and anchor positions on localization. For the filter-based approach [6], [7], the tag location as well as the tag velocity or the bearing information are used to establish the state equations. Besides, the ranging measurements are utilized as the observation. Then, the state variables can be estimated based on the Kalman filter (KF) framework. However, the linearization process of the KF will bring extra error to the system. For the optimization-based approach, a function to be optimized is established instead of building the state and observation equations. In [8], the measurement noise is fully considered in optimal function, while the maximum likelihood estimation approach is used to estimate the tag position. In [9], rather than the ranging measurement constraints, it introduces the maximum speed of the tag into the optimal function as the constraints between consecutive frames, leading to the state-of-the-art UWB localization performance in classical approaches. The complete objective function is defined as follows [9], and it can be optimized by a nonlinear optimization method, such as Gauss–Newton or Levenberg–Marquardt

$$\boldsymbol{F}_{\boldsymbol{UWB}} = \min_{\hat{p}} \sum_t \Bigg(W_{t,t-1} \left(\left\| \widehat{\boldsymbol{p}}_t - \widehat{\boldsymbol{p}}_{t-1} \right\|_2 \right)^2 + \sum_i W_{t,i} \left(\hat{d}_{t,i} - \left\| \widehat{\boldsymbol{p}}_t - \boldsymbol{A}_i \right\|_2 \right)^2 \Bigg) \tag{2}$$

where $W_{t,i}$ is the weight parameter depending on the sensor precision, while $W_{t,t-1}$ is another weight parameter according to the maximum speed of the tag.

Recently, many works attempt to utilize various deep neural networks to further improve the localization performance in

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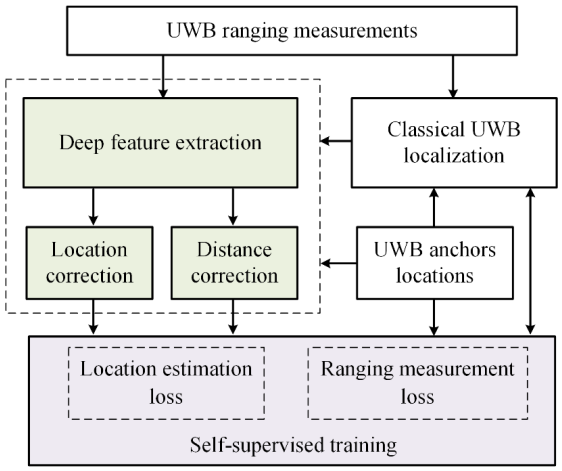


Fig. 1. Flowchart of our proposed self-supervised deep UWB localization method: our proposed DLRC network (green blocks) regresses location and ranging error corrections to a classical UWB localization result, while a self-supervised learning strategy (purple block) is designed to train the whole network.

complex environments. In [12], a convolutional neural network (CNN) is first used to handle the measured distance and compute the tag location in an end-to-end manner. The long short-term memory (LSTM) [13] as well as the gate recurrent unit (GRU) [14] are also introduced to establish the temporal relation of the UWB data, and then estimate the localization results. In our previous work [15], we jointly learn the spatial-temporal features from the UWB measurements and achieve the desirable localization performance in the non-line of sight (NLOS) scenes. Compared to the classical approaches, the greatest advantages of the deep learning-based method are: 1) it is free from the positions of UWB anchors due to its training process with ground truth and 2) it can learn high-level representations of the measurements, and thus obtain accurate results based on this representation. However, all these existing UWB localization deep networks are trained with ground truth which is costly to collect in practice. Thus, it limits the utilization of deep learning-based methods. This motivates us to propose a self-supervised learning strategy.

In addition, in our method, we need to estimate the ranging error corrections. In terms of the ranging error mitigation approaches, many existing works are proposed based on the machine learning approach [16], [17], [18]. The sparse pseudo-input Gaussian process [16], semi-definite relaxation technique [17], and support vector machine with a semi-supervised approach [18] are successively proposed to correct the ranging measurements by using the channel impulse responses (CIR) waveforms or received signal strength (RSS) information. In addition, the CNN is also introduced to estimate the distance for UWB localization to improve the localization accuracy [19]. However, the CIR and RSS information widely used in the ranging error mitigation are not considered in this article.

III. Proposed Methods

A. System Overview

In this article, our purpose is to correct the ranging measurements and tag locations by building a deep network with a self-supervised training manner. Thus, as shown in Fig. 1 which illustrates the flowchart of our proposed method, we first utilize a classical UWB localization approach (e.g., least squares-based or optimization-based approach) to provide initial localization results. Then, the following two parts are presented to correct the initial tag position and the raw distances measured with the self-supervised training manner.

1) DLRC network is designed to regress location and ranging error corrections to the classical UWB localization results. Specifically, it includes three modules: the feature extraction module is constructed using convolutional layers to extract the deep spatial features from the UWB ranging measurements. The location correction module and distance correction module are designed to estimate the location and ranging error corrections based on the extracted features, initial tag positions, and anchor locations.
2) Self-supervised learning strategy is proposed to train the DLRC network without involving the ground truth. In this training strategy, we mainly build location estimation loss and ranging measurements loss. They can constrain the location and ranging error corrections by considering the localization results from classical approach and known positions of UWB anchors. Thus, the proposed network can be trained in a self-supervised manner.

B. Model Architecture

Fig. 2 shows the architecture of our proposed DLRC network. First, the UWB ranging measurements are collected as the input of the feature extraction module (block 1 in Fig. 2). To be specific, in order to contain more useful information, ten consecutive ranging measurements between the tag and one anchor are utilized as a sample in our network. Thus, the input data are defined as follows:

$$X_{t,i} = (\hat{d}_{n-9,i}, \hat{d}_{n-8,i}, \ldots, \hat{d}_{n,i}), \quad i \in [1, 2, \ldots, M] \tag{3}$$

where $X_{t,i}$ is one sample between the tag and the ith anchor at frame t. $\hat{d}_{n,i}$ is the ranging measurement between the tag and the ith anchor at the UWB measurement time n. M is the number of anchors.

There will be M samples at a frame for network input. Besides, average values of samples $\hat{d}_{t,i}$ will be used for the classical UWB localization approach to compute the initial tag location.

Next, we construct the feature extraction module with five convolutional layers (block 2 in Fig. 2). Then, the deep spatial features of samples (block 3 in Fig. 2) can be obtained as follows:

$$\mathrm{FeaU}_{t,i} = \Phi\left(X_{t,i}\right), \quad i \in [1, 2, \ldots, M] \tag{4}$$

where $\mathrm{FeaU}_{t,i}$ is the extracted deep spatial features of ith samples at frame t. $\Phi(\bullet)$ represents the feature extraction module. It maps the $X_{t,i}$ to $\mathrm{FeaU}_{t,i}$.

The details of the feature extraction module are provided in Table I. When the dimension of input sample is 10, we use 1-D convolution operation to handle this kind of data. First, a wide kernel with size 5 is used to suppress the high-frequency

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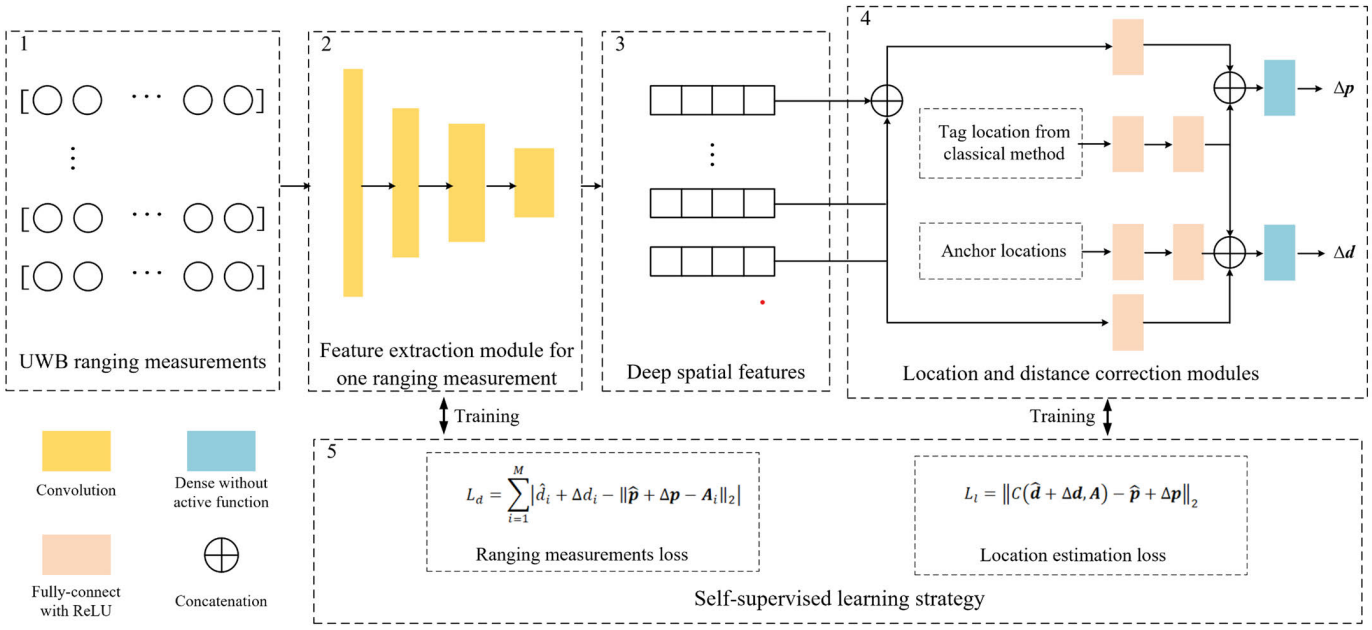


Fig. 2. Network diagram of our self-supervised DLRC network: our network jointly estimates a location correction $\Delta \boldsymbol{p}$ and a distance correlation $\Delta \boldsymbol{d}$. Besides, we present a self-supervised learning strategy based on two main losses: ranging measurements loss and location estimation loss, where $\widehat{\boldsymbol{d}}$ is the UWB ranging measurements, $\widehat{\boldsymbol{p}}$ is the tag location from classical method, and $\boldsymbol{A}$ is the anchors locations.

TABLE I
DETAILS OF FEATURE EXTRACTION MODULE WITH FIVE CONVOLUTIONAL LAYERS

Layer Type	Kernel Size	Kernel Number	Output Sizes
Convolution 1	5	32	32×6
Convolution 2	3	64	64×4
Convolution 3	2	64	64×3
Convolution 4	2	128	128×2
Convolution 5	2	128	128×1

measurement noise of the UWB sensor, then relatively small kernels with sizes 3 and 2 are selected to make the network deeper, leading to the representative features of the ranging measurements. For each convolutional layer, ReLU is used as the activation function. Meanwhile, batch normalization [20] is introduced to prevent overfitting and accelerate the training. We avoid utilizing pooling layers and padding operations as they will reduce the distinctiveness of the features or introduce additional noise. After the fifth convolutional layers, the output of the feature extraction module is a vector with 128 elements.

Finally, the location and ranging error corrections can be estimated based on the deep spatial features, the initial tag location, and anchors positions (block 4 in Fig. 2). This process contains three steps: 1) we extract the features of initial localization results and anchors positions, respectively, as follows:

$$\mathrm{FeaL}_t = \mathrm{FC}_2\left(C\left(\boldsymbol{d}_t, \boldsymbol{A}\right)\right) \tag{5a}$$

$$\mathrm{FeaA}_i = \mathrm{FC}_2\left(\boldsymbol{A}_i\right), \quad i \in [1, 2, \ldots, M] \tag{5b}$$

where FeaL_t is the deep features of initial localization results. FeaA_i is the deep features of ith anchor position. $C(\boldsymbol{d}_t, \boldsymbol{A})$ means the localization results from classical UWB localization approach with distance information $\boldsymbol{d}_t$ and fixed anchor locations $\boldsymbol{A}$. $\boldsymbol{A}_i$ is the fixed location of ith anchor. $\mathrm{FC}_2(\bullet)$ represents two fully connected layers with ReLU function.

These equations show that two independent submodules are designed to generate the features of initial localization results and features of anchors' positions, respectively. These two submodules include two fully connected layers with the same dimensions, but their parameters are not shared. The input dimension of their first fully connected layers is set as 3 illustrating the dimension of tag or anchors positions, while the output dimension of these two layers is, respectively, 128 and 64.

2) The location corrections $\Delta \boldsymbol{p}_t$ can be computed as follows:

$$\Delta \boldsymbol{p}_t = D\left(\mathrm{FC}_1\left(\mathrm{FeaU}_{t,1} \oplus \mathrm{FeaU}_{t,2} \oplus, \ldots, \mathrm{FeaU}_{t,M}\right) \oplus \mathrm{FeaL}_t\right) \tag{6}$$

where $\mathrm{FC}_1(\bullet)$ represents one fully connected layer with ReLU function. $D(\bullet)$ represents one dense layer without active functions. $\oplus$ is the connection operation.

Specifically, the M features of ranging measurements are concatenated first and then combined through a fully connected layer with the 64 output dimensions. Next, this output is concatenated with the features of initial tag locations and then regressed to the location corrections by a dense layer with 32 and 3 output dimensions, respectively.

3) The ranging error corrections $\Delta d_{t,i}$ can be calculated as follows:

$$\Delta d_{t,i} = D\left(\mathrm{FeaL}_t \oplus \mathrm{FeaA}_i \oplus \mathrm{FC}_1\left(\mathrm{FeaU}_{t,i}\right)\right), \quad i \in [1, 2, \ldots, M]. \tag{7}$$

To be specific, we use a fully connected layer with output dimension 64 to handle each ranging measurement feature.

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Subsequently, for each handled ranging measurement feature, we concatenate it with the feature of initial tag location and feature of the relative anchor position. Then, each ranging error correction is computed based on the concatenated features by using a dense layer with 32 and 1 output dimensions, respectively.

Besides, the dropout [21] is also introduced to every fully connected and dense layer to avoid overfitting problem of network training.

In addition, we discuss the case when the signal of several UWB anchors is lost. In this situation, less than M ranging measurements can be provided in one time step. For the training data, we use the linear interpolation method to fill in the missing measurements. In the testing process, we use the vectors with 128 zero elements as their features to guarantee dimension of the concatenated ranging measurement features is $128M$. Besides, if the available measured distances cannot support the classical approach to compute an initial tag location in a frame, we set this initial location as the corrected tag location in the previous frame to guarantee the localization continuity. However, in the cases when most of the anchors are unavailable for a long time, the accuracy of the localization results will decline.

C. Self-Supervised Learning Strategy

The topological structure of the UWB sensors is mainly contained in the positions of the UWB anchors. In order to introduce the topological information of the UWB sensors into the DLRC network and train the network without the ground truth, we present a self-supervised learning strategy based on two main losses: ranging measurements loss and location estimation loss (block 5 in Fig. 2). We now give the details of these two losses.

At frame t, the proposed DLRC network generates the location correction $\Delta \boldsymbol{p}_t$ and the ranging error correction $\Delta \boldsymbol{d}_t$, while the classical UWB localization approach obtains the estimated tag location $\widehat{\boldsymbol{p}}_t$, and the UWB sensor measures the distance information $\widehat{\boldsymbol{d}}_t$ between the tag and anchors. We can use the $\Delta \boldsymbol{p}_t$ and $\Delta \boldsymbol{d}_t$ to correct the estimated tag location $\widehat{\boldsymbol{p}}_t$ and the measured distance $\widehat{\boldsymbol{d}}_t$ to achieve a more accurate tag location $\boldsymbol{p}_t$ and distance information $\boldsymbol{d}_t$ as follows:

$$\boldsymbol{p}_t = \widehat{\boldsymbol{p}}_t + \Delta \boldsymbol{p}_t \tag{8a}$$

$$\boldsymbol{d}_t = \widehat{\boldsymbol{d}}_t + \Delta \boldsymbol{d}_t. \tag{9a}$$

For $\boldsymbol{p}_t$ and $\boldsymbol{d}_t$, if they are exact tag location and the distances between tag and anchors, the distances between $\boldsymbol{p}_t$ and anchors will equal to $\boldsymbol{d}_t$. Thus, the ranging measurements loss can be constructed as follows:

$$L_d = \sum_{i=1}^{M} \left| d_{t,i} - \left\| \boldsymbol{p}_t - \boldsymbol{A}_i \right\|_2 \right| \tag{10}$$

where $\boldsymbol{A}_i$ is the fixed location of the ith anchor, $\|\boldsymbol{p}_t - \boldsymbol{A}_i\|_2$ is the distance between $\boldsymbol{p}_t$ and the ith anchor, $d_{t,i}$ is the corrected distance between the tag and the ith anchor, and M is the number of anchors.

Similarly, if $\boldsymbol{p}_t$ and $\boldsymbol{d}_t$ are exactly estimated, the classical UWB localization approach with $\boldsymbol{d}_t$ can calculate an accurate tag location which is equal to $\boldsymbol{p}_t$. Therefore, the location estimation loss can be built as follows:

$$L_l = \left\| C\left(\boldsymbol{d}_t, \boldsymbol{A}\right) - \boldsymbol{p}_t \right\|_2 \tag{11}$$

where $C(\boldsymbol{d}_t, \boldsymbol{A})$ means the localization results from classical UWB localization approach with distance information $\boldsymbol{d}_t$ and fixed anchor locations $\boldsymbol{A}$.

Finally, the whole loss function for training is defined as follows:

$$L = \gamma_1 L_d + \gamma_2 L_l \tag{12}$$

where γ_1 and γ_2 are weight parameters of ranging measurements and location estimation loss. These two weight parameters can be set empirically according to the environment. Generally, γ_1 can be set around the 1.0. In terms of γ_2, for an ideal environment with small-ranging errors, it can be set close to γ_1. However, for a limited environment with large-ranging noise, it should be set to a small value such as 0.2. Besides, a slightly value change of these two parameters will not affect the accuracy of the proposed method.

By using this loss function, on the one hand, the ground truth is not required. Thus, the proposed network can be trained in a self-supervised manner. On the other hand, in this loss function, the location correction and the ranging error correction are constrained jointly by the precalculated tag location derived from the classical approach and the known anchors' positions which contain the topological information of UWB sensors. Hence, the geometric constraints can be introduced into the DLRC network through it.

IV. Evaluation

A. Experimental Settings

In this section, we build real-world experiments in two different environments with different UWB devices to evaluate the proposed UWB localization method comprehensively. As shown in Figs. 3 and 4, experimental field 1 is a 6 × 5 m laboratory environment, while experimental field 2 is a 10 × 5 m apartment scene. We both install four UWB anchors in these two environments. The anchor positions in experimental field 1 are (−0.15, −0.31, 2.20), (−0.21, 2.33, 1.70), (4.04, 2.64, 1.80), and (5.74, −0.08, 1.50), while they are (0, 0, 0.995), (7.2, −0.35, 1.01), (0, 2.2, 2.02), and (6.4, 1.48, 1.96) in experimental field 2.

In terms of UWB devices, we use Mini3sPlus for experimental field 1 and LinckTrack-S for experimental field 2. Both of them are based on DWM 1000 chips which can provide the distance information between tag and anchors within 80 m. Owing to the situation of different environments, the average ranging accuracy in the laboratory environment is less than 10 cm, while it is around 16 cm in the apartment scene. The measurement frequency of the LinckTrack-S is 100 Hz, and it is 10 Hz for Mini3sPlus. Consequently, for the experiments in field 1, we need to augment the UWB ranging measurement to 100 Hz by using a linear interpolation approach. Finally, these two UWB devices are low-cost (<U.S. $100 each anchor or tag) and are affordable in practice.

To train and test the proposed method, we build a mobile robot with NVIDIA AGX Xavier, a 32-line LiDAR

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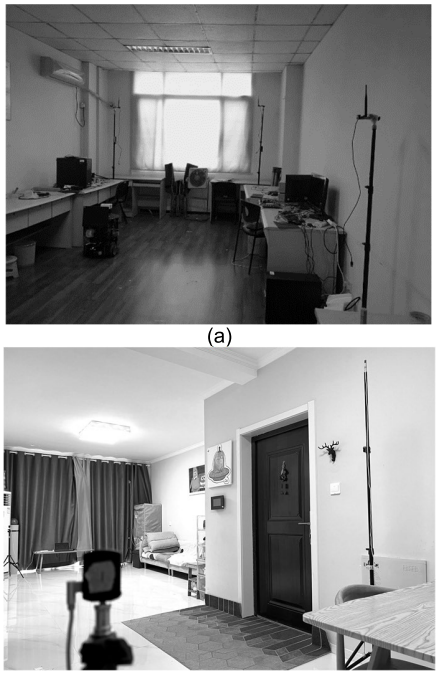


Fig. 3. Images of (a) laboratory environment and (b) apartment scene.

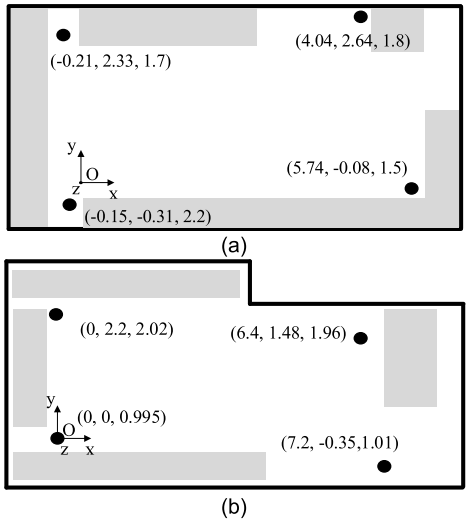


Fig. 4. Structure map of (a) laboratory environment and (b) apartment scene. The black lines indicate the walls, while the gray rectangles illustrate the furniture. Besides, the black points represent the anchor positions.

(RS-Helios-1615), as well as the corresponding UWB tag. We use this robot to collect the training data, and real-time testing is also based on this platform. In addition, the 3-D LiDAR with the stable and cm-level localization method LOAM [22] is used to provide the ground truth for testing.

We use absolute trajectory error (ATE) [23] of the location as the metric to evaluate the localization accuracy. In terms of comparison approaches, the least squares-based method [5] and the state-of-the-art optimization-based method [9] are used as classical approaches. Since our proposed method aims to correct the results of the classical approaches, we mainly compare the performance of our method with these classical approaches to show the effectiveness of our method. Besides, we also give the discussion of our proposed method and the state-of-the-art supervised learning method [16].

TABLE II
SITUATION OF EACH SEQUENCE IN THE LABORATORY ENVIRONMENT

Sequence	Type	Number of samples
Seq 2,4,5,6,9,10,12,13,15,16,17,19,20,22	Train	31306
Seq 1	Test	2622
Seq 3	Test	2444
Seq 7	Test	1391
Seq 8	Test	2248
Seq 11	Test	1670
Seq 14	Test	1897
Seq 18	Test	1750
Seq 21	Test	1968

TABLE III
ATE OF TESTING SEQUENCES IN THE LABORATORY SCENE WITH LEAST SQUARES-BASED METHOD AS CLASSICAL APPROACH (*M*)

	Least squares-based [5]				Our (least squares-based method as classical approach)			
	x	y	z	total	x	y	z	total
Seq 1	0.056	0.051	0.564	0.569	0.050	0.046	0.221	0.232
Seq 3	0.063	0.049	0.529	0.535	0.064	0.048	0.248	0.260
Seq 7	0.081	0.043	0.484	0.492	0.067	0.048	0.142	0.163
Seq 8	0.069	0.046	0.440	0.447	0.059	0.046	0.154	0.171
Seq 11	0.074	0.045	0.601	0.607	0.063	0.039	0.218	0.230
Seq 14	0.073	0.050	0.465	0.473	0.068	0.048	0.150	0.172
Seq 18	0.068	0.042	0.493	0.499	0.064	0.039	0.155	0.173
Seq 21	0.067	0.045	0.591	0.597	0.065	0.042	0.225	0.238

In terms of the training setup, we first collect enough UWB ranging measurements in two environments and precalculate the estimated tag location by classical localization approaches. In the training process, these collected ranging measurements are fed into the DLRC network as well as the estimated tag location. We use the Adam [24] as the optimizer owing to its high-quality performance. The weight-decay of the Adam is set as 0.0001 to further prevent overfitting, while the learning rate is set as 0.0001, the batch size is set as 1024, and γ_1 and γ_2 are empirically set as 1.0 and 0.2, respectively. In addition, the whole system is implemented based on Python, while the DLRC network is implemented using Pytorch, and trained on a NVIDIA GeForce RTX 3060.

B. Experimental Results in the Laboratory Environment

In this experiment, we control the mobile robot to walk randomly in the laboratory scene and record 22 sequences of data. We randomly divide these sequences into 14 training sequences and eight testing sequences. The details of these data are shown in Table II.

In terms of the classical approaches, the least squares-based method [5] and the state-of-the-art optimization-based method [9] are used, respectively, for testing. Meanwhile, these two methods are used for our proposed method, respectively, to provide initial tag positions.

The results of all testing sequences of least squares-based method and our proposed method with least squares-based method as classical approach are shown in Table III. In this table, the total ATE of tag positions and the ATE of tag

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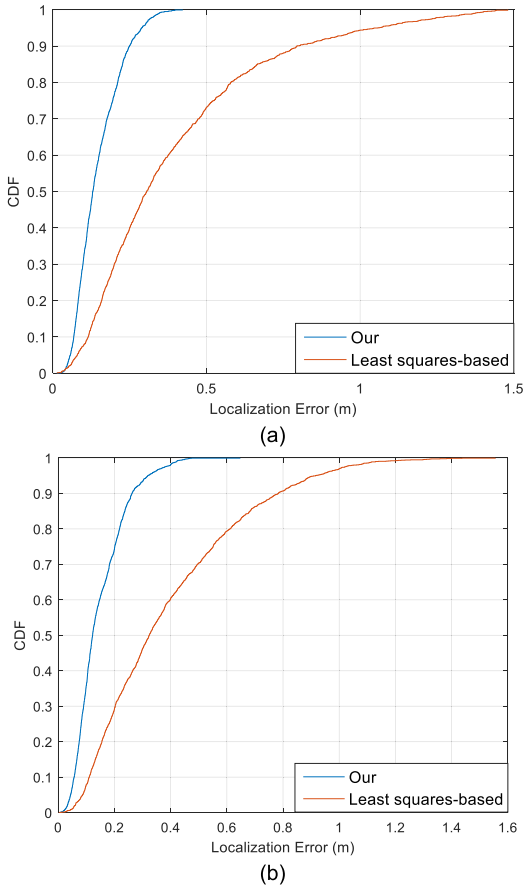


Fig. 5. CDF of localization errors of (a) sequence 7 and (b) sequence 14 of least squares-based method and our proposed method with least squares-based method as the classical approach. They show that our proposed method beats the least squares-based method in these sequences.

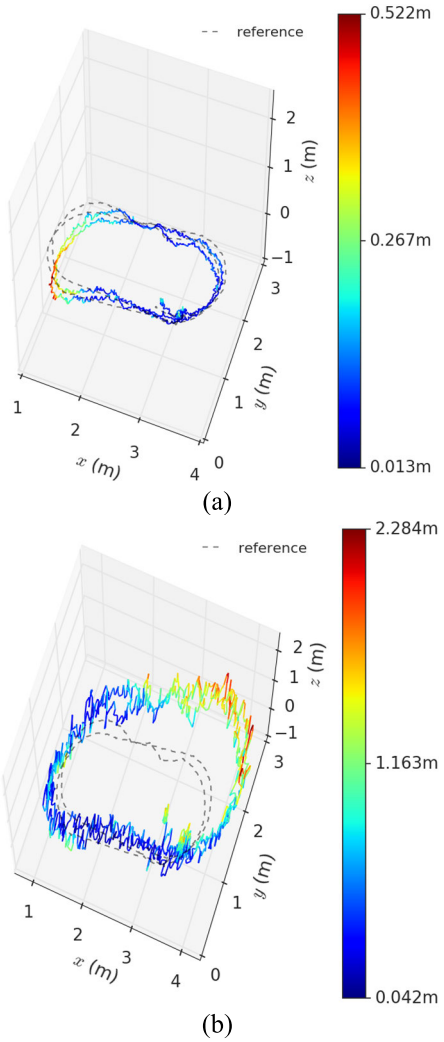


Fig. 6. Trajectory error curves of (a) our proposed method with least squares-based method as the classical method and (b) least squares-based method of sequence 18. It shows that the trajectory errors of least squares-based method mainly exist in z-axis. Our proposed DLRC network can significantly improve its localization accuracy of z-axis, and also can correct the localization results of x-axis and y-axis reporting desirable localization accuracy.

positions in three axes are all given. We also provide the cumulative distribution function (cdf) curves and trajectory error curves of these two methods in Fig. 5. Overall, these results suggest that our proposed method can correct the tag position provided by least squares-based method and obtain better localization performance.

To be specific, Table III shows that the least squares-based method achieves the worst results because its accuracy of the z-axis is lowest caused by the anchor positions (the coordinate differences of z-axis are smaller than others). This situation can also be observed in Fig. 6(b) where the trajectory errors of least squares-based method mainly exist in z-axis. By using these results for our proposed method as the initial tag positions, our proposed DLRC network can significantly improve its localization accuracy of z-axis, and also can correct the localization results of x-axis and y-axis reporting desirable localization accuracy. Figs. 5 and 6(a) also illustrate these improvements clearly through the specific sequences.

Similarly, the results of all testing sequences of optimization-based method and our proposed method with optimization-based method as classical approach are shown in Table IV. This table shows that the optimization-based method achieves a high-quality performance in this experiment due to the strong optimization function, whereas our developed approach can also correct its localization results in all axes, and achieve the performance improvements when using this method as the classical approach. We also provide several cdf curves and trajectory error curves of specific sequences in Figs. 7 and 8. Fig. 7 shows that the cdf of our proposed

TABLE IV
ATE OF TESTING SEQUENCES IN THE LABORATORY SCENE WITH OPTIMIZATION-BASED METHOD AS CLASSICAL APPROACH (M)

	Optimization-based [9]				Our (optimization-based method as classical approach)			
	x	y	z	total	x	y	z	total
Seq 1	0.058	0.069	0.061	0.110	0.051	0.060	0.057	0.098
Seq 3	0.060	0.057	0.065	0.106	0.052	0.055	0.055	0.093
Seq 7	0.076	0.031	0.096	0.127	0.063	0.028	0.079	0.105
Seq 8	0.071	0.046	0.078	0.115	0.073	0.043	0.065	0.107
Seq 11	0.070	0.049	0.079	0.116	0.064	0.041	0.069	0.102
Seq 14	0.073	0.037	0.097	0.127	0.063	0.031	0.076	0.103
Seq 18	0.072	0.036	0.138	0.160	0.065	0.037	0.079	0.109
Seq 21	0.068	0.051	0.072	0.111	0.062	0.051	0.061	0.101

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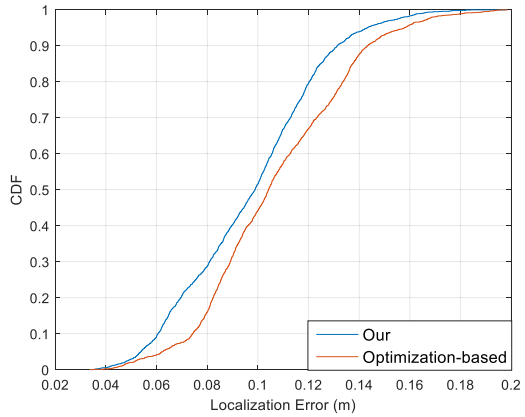


Fig. 7. CDF of localization errors of sequence 21 of optimization-based method and our proposed method with optimization-based method as the classical approach. It shows that our proposed method beats the optimization-based method in this sequence.

TABLE V
AVERAGE DISTANCE ERROR OF TESTING SEQUENCES IN THE LABORATORY ENVIRONMENT(*M*)

	Raw measurements	Our (least squares-based method as classical approach)	Our (optimization-based method as classical approach)
Seq 1	0.081	0.057	0.046
Seq 3	0.076	0.056	0.047
Seq 7	0.074	0.052	0.045
Seq 8	0.077	0.051	0.050
Seq 11	0.088	0.068	0.057
Seq 14	0.074	0.054	0.052
Seq 18	0.078	0.050	0.046
Seq 21	0.088	0.064	0.052

method is always larger than optimization-based method, suggesting the better localization performance of our DLRC network. Besides, by comparing Fig. 8(a) and (b) which illustrate the trajectory errors of our proposed method and optimization-based method of sequence 18, respectively, it shows clearly that our DLRC network achieves more accurate localization results.

These experimental results indicate the effectiveness of our DLRC network and our self-supervised learning strategy. In addition, from Tables III and IV, we can observe that the DLRC network obtains better performance when the optimization-based method is used as the classical approach to provide the initial tag positions, suggesting that more accurate classical approach leads to better localization results.

Finally, the distance error of the raw measurements and the outputs of the DLRC network are given in Table V. It shows that the DLRC network can also correct the ranging measurements effectively for the classical approaches. Besides, better correction performance is revealed when using high-accurate classical approaches.

C. Experimental Results in the Apartment Environment

In terms of the experiments in the apartment environment, we control the mobile robot to collect two sequences of data for training, and then we construct real-time testing experiments based on the mobile robot. The LinckTrack-S

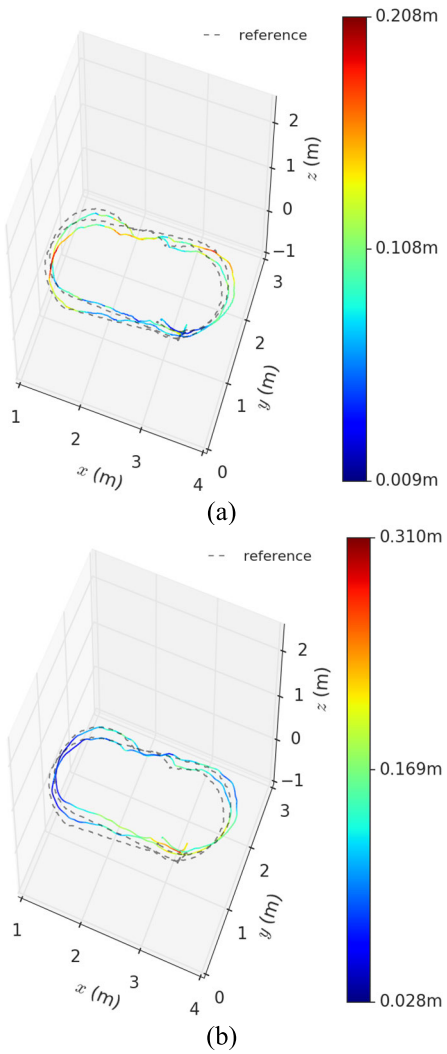


Fig. 8. Trajectory error curves of (a) our proposed method with optimization-based method as the classical method and (b) optimization-based method of sequence 18. Our DLRC network achieves more accurate localization results since the maximum, medium, and minimum trajectory errors of our proposed method are all better than the optimization-based method.

TABLE VI
SITUATION OF EACH SEQUENCE IN THE APARTMENT ENVIRONMENT

Sequence	Type	Number of samples
Seq 0,1	Train	19235
Seq 2	Test	1389
Seq 3	Test	2942
Seq 4	Test	2315
Seq 5	Test	2694
Seq 6	Test	2204
Seq 7	Test	2759
Seq 8	Test	1629

UWB devices measure the distance information in real-time; meanwhile, the proposed method is carried out to compute the localization results with the NVIDIA AGX Xavier. The details of these sequences are shown in Table VI.

In this experiment, due to the unfriendly anchor positions and the relatively large-ranging measurements, the

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TABLE VII
ATE OF TESTING SEQUENCES IN THE APARTMENT SCENE WITH OPTIMIZATION-BASED METHOD AS CLASSICAL APPROACH (M)

	Optimization-based [9]				Our (optimization-based method as classical approach)			
	x	y	z	total	x	y	z	total
Seq 2	0.073	0.243	0.373	0.451	0.072	0.103	0.138	0.187
Seq 3	0.099	0.251	0.356	0.447	0.091	0.131	0.127	0.204
Seq 4	0.100	0.283	0.369	0.476	0.092	0.164	0.125	0.226
Seq 5	0.100	0.265	0.392	0.484	0.107	0.171	0.160	0.258
Seq 6	0.093	0.275	0.384	0.481	0.091	0.132	0.155	0.224
Seq 7	0.096	0.273	0.384	0.481	0.094	0.162	0.147	0.237
Seq 8	0.086	0.231	0.382	0.454	0.086	0.106	0.138	0.195

TABLE VIII
AVERAGE DISTANCE ERROR OF TESTING SEQUENCES IN THE APARTMENT ENVIRONMENT (M)

	Raw measurements	Our (optimization-based method as classical approach)
Seq 2	0.149	0.089
Seq 3	0.152	0.103
Seq 4	0.156	0.110
Seq 5	0.162	0.121
Seq 6	0.157	0.114
Seq 7	0.164	0.111
Seq 8	0.160	0.093

least squares-based method fails in this environment. Hence, we only use the optimization-based method to provide the initial tag positions for our designed DLRC network in this scene.

Table VII illustrates the localization results of our proposed method and optimization-based method of all testing sequences in the apartment environment. It shows that even the state-of-the-art optimization-based method cannot achieve a desirable localization performance in this more complex scene. However, the proposed DLRC network corrects these localization biases successfully in all three axes and obtains the performance improvement. In order to show this improvement intuitively, we give the cdf curves of sequence 3 and sequence 8 of these two methods in Fig. 9. The figure shows that the cdfs of our proposed method are always larger than the state-of-the-art optimization-based method significantly indicating that our proposed method can improve the localization accuracy compared to optimization-based method. In addition, trajectory error curves of these two methods of sequence 2 are given in Fig. 10. It shows that the maximum, medium, and minimum trajectory errors of our proposed method are all better than the optimization-based method significantly which is similar to the above results. This result suggests the better localization performance of our proposed method.

Finally, the distance error of the raw measurements and the outputs of the DLRC network are provided in Table VIII. It also shows that our proposed DLRC network can output more accurate distance information between the tag and anchors than the ranging measurements.

The above two experiments clearly demonstrate the advantages of our proposed DLRC network and the self-supervised learning strategy. Our proposed self-supervised DLRC network can correct the tag positions provided by the classical

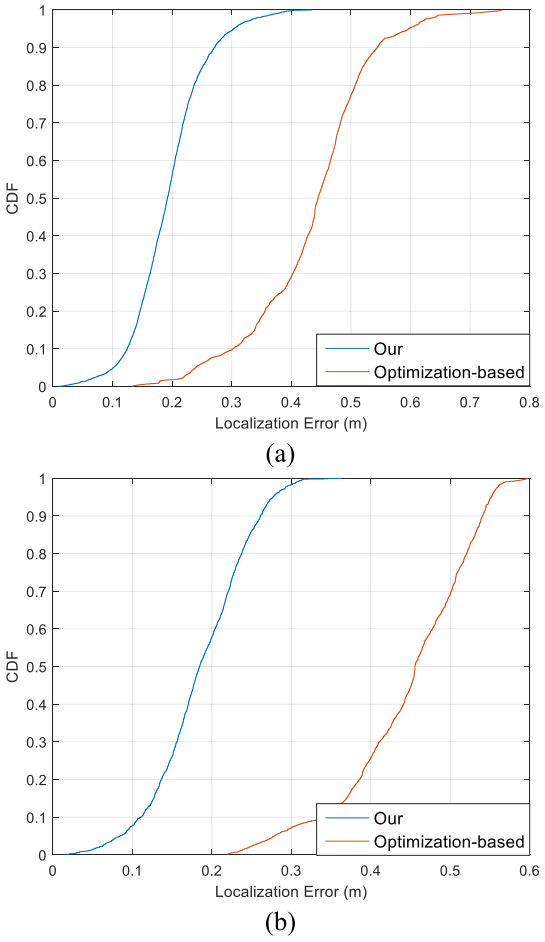


Fig. 9. CDF of localization errors of (a) sequence 3 and (b) sequence 8 of optimization-based method and our proposed method with optimization-based method as the classical approach. They show that our proposed method achieves better results than the optimization-based method significantly in these sequences.

method achieving better localization performance. Meanwhile, it can also correct raw ranging measurements obtaining more accurate distance information. Besides, the training process is free from the ground truth which is convenient in applications.

In addition, the initial tag positions provided by the classical methods also affect the accuracy of our proposed method. More accurate initial tag positions lead to better localization results. Thus, in an environment where the ranging measurement noise is very large, the state-of-the-art classical methods even cannot provide available localization results. As a result, our proposed method also cannot achieve reliable performance.

Finally, we discuss the cost of our proposed method. Although the proposed method takes some time to train the DLRC network, the training process is independent of ground truth, and thus it is easy to deploy in practice. Besides, the average time cost of the DLRC network is about 0.002 s/frame which will not influence the localization frequency.

D. Self-Supervised Versus Supervised Learning

Finally, we give the localization results of the above two experiments of the state-of-the-art supervised learning method [15] in Tables IX and X.

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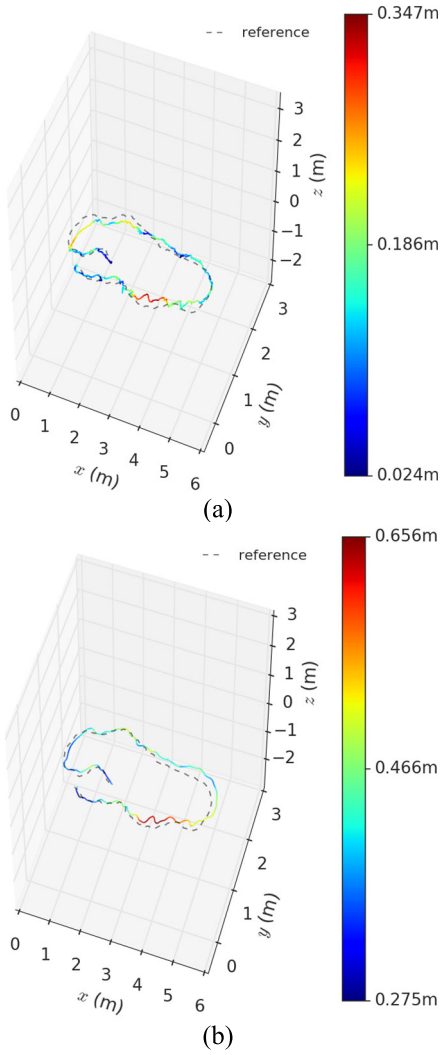


Fig. 10. Trajectory error curves of (a) our proposed method with optimization-based method as the classical method and (b) optimization-based method of sequence 2.

TABLE IX
ATE OF SUPERVISED LEARNING METHOD AND OUR PROPOSED METHOD (OPTIMIZATION-BASED METHOD AS CLASSICAL APPROACH) OF TESTING SEQUENCES IN THE LABORATORY SCENE (*M*)

	Supervised learning method [15]				Our (optimization-based method as classical approach)			
	x	y	z	total	x	y	z	total
Seq 1	0.050	0.055	0.029	0.080	0.051	0.060	0.057	0.098
Seq 3	0.061	0.053	0.023	0.084	0.052	0.055	0.055	0.093
Seq 7	0.065	0.069	0.026	0.099	0.063	0.028	0.079	0.105
Seq 8	0.053	0.052	0.019	0.077	0.073	0.043	0.065	0.107
Seq 11	0.053	0.045	0.022	0.073	0.064	0.041	0.069	0.102
Seq 14	0.049	0.053	0.017	0.076	0.063	0.031	0.076	0.103
Seq 18	0.051	0.031	0.021	0.063	0.065	0.037	0.079	0.109
Seq 21	0.053	0.042	0.017	0.069	0.062	0.051	0.061	0.101

Overall, the state-of-the-art supervised learning method can achieve better performances than our proposed self-supervised method. However, these accuracy improvements are mainly from the z-axis because the moving distance of the robot on z-axis is small, thus the supervised learning method

TABLE X
ATE OF SUPERVISED LEARNING METHOD AND OUR PROPOSED METHOD (OPTIMIZATION-BASED METHOD AS CLASSICAL APPROACH) OF TESTING SEQUENCES IN THE APARTMENT SCENE (*M*)

	Supervised learning method [15]				Our (optimization-based method as classical approach)			
	x	y	z	total	x	y	z	total
Seq 2	0.089	0.107	0.029	0.143	0.072	0.103	0.138	0.187
Seq 3	0.103	0.111	0.034	0.156	0.091	0.131	0.127	0.204
Seq 4	0.118	0.122	0.033	0.173	0.092	0.164	0.125	0.226
Seq 5	0.115	0.129	0.043	0.178	0.107	0.171	0.160	0.258
Seq 6	0.107	0.117	0.036	0.163	0.091	0.132	0.155	0.224
Seq 7	0.108	0.113	0.042	0.162	0.094	0.162	0.147	0.237
Seq 8	0.102	0.095	0.026	0.141	0.086	0.106	0.138	0.195

can regress the location of z-axis accurately based on the precollected ground truth. In terms of the other axis, our proposed method can achieve comparable results with the supervised learning method since the geometric constraints are considered in our method through the self-training strategy and the introduction of classical approach. In addition, as mentioned before, our proposed method is free from the ground-truth collection which reduces the training cost and it is convenient for the practices when there are no devices that can provide the ground truth.

V. CONCLUSION AND FUTURE WORK

In this article, we propose a self-supervised DLRC method. This method can improve the localization accuracy compared to state-of-the-art classical approaches with desirable computational complexity. In this method, we fuse the classical approach with the self-supervised learning-based method through the estimation of the location and ranging error corrections with designed deep network. The greatest advantage of our method is the self-supervised learning strategy based on the topological structure of UWB sensors. This strategy removes the requirement of ground-truth collection, and thus our method can use the deep measurement features without the ground truth. Consequently, our approach facilitates real-world applications and can achieve a desirable localization result. The advantages of the proposed method are also suggested through the real-world experiments with different UWB devices in different environments.

In future work, we will consider designing a more powerful feature extraction module to further extract higher quality features from the UWB measurements. In addition, other sensors such as inertial measurement unit will be considered to integrate with the UWB through the proposed self-supervised deep learning-based framework.

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